Experiment 10: Performance Optimization and Best Practices

Aim

To keep the gym application fast and easy to maintain as the number of members grows.

Theory

Performance depends on how forms, reports, workflows and scripts are designed. Reports that load every column, scripts that fetch every record, and repeated typing of the same data slow an application.

Procedure

1. Used a Lookup to the Trainers form instead of typing trainer names.
2. Limited the SubForm Family_Members_Enrolled to 5 rows.
3. Added only the needed columns to the report (Membership ID, QR code, total, trainer).
4. Kept the validation script short, with simple if conditions and no record fetches.
5. Used a dashboard with count and chart so that totals are not calculated by hand.

Output

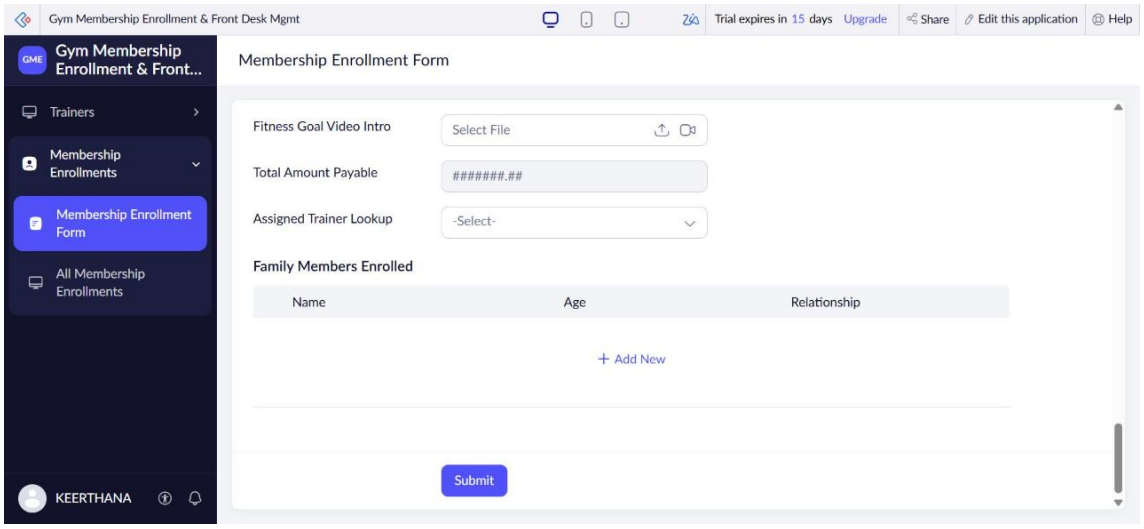


Fig 10.1 Lookup and limited SubForm in the form

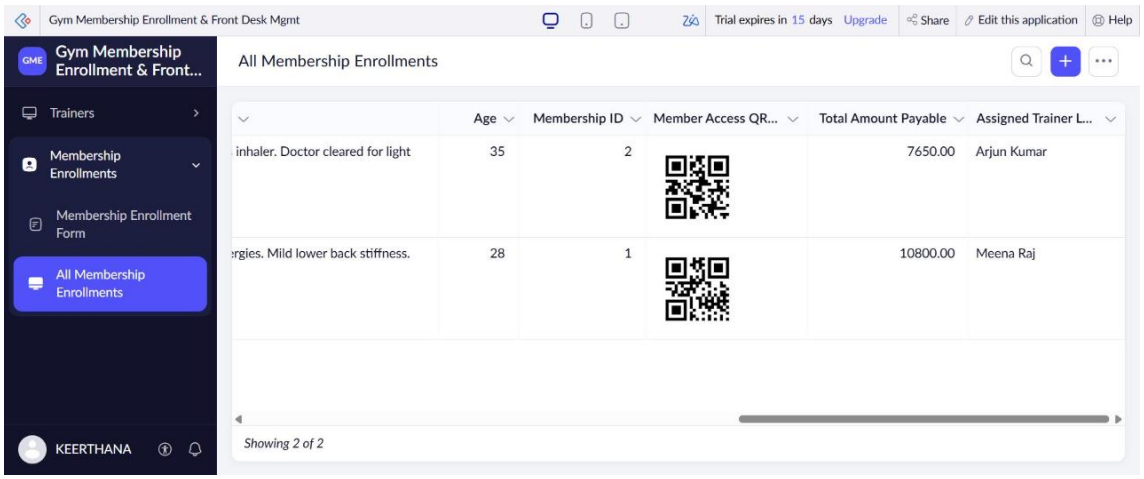


Fig 10.2 Report with selected columns

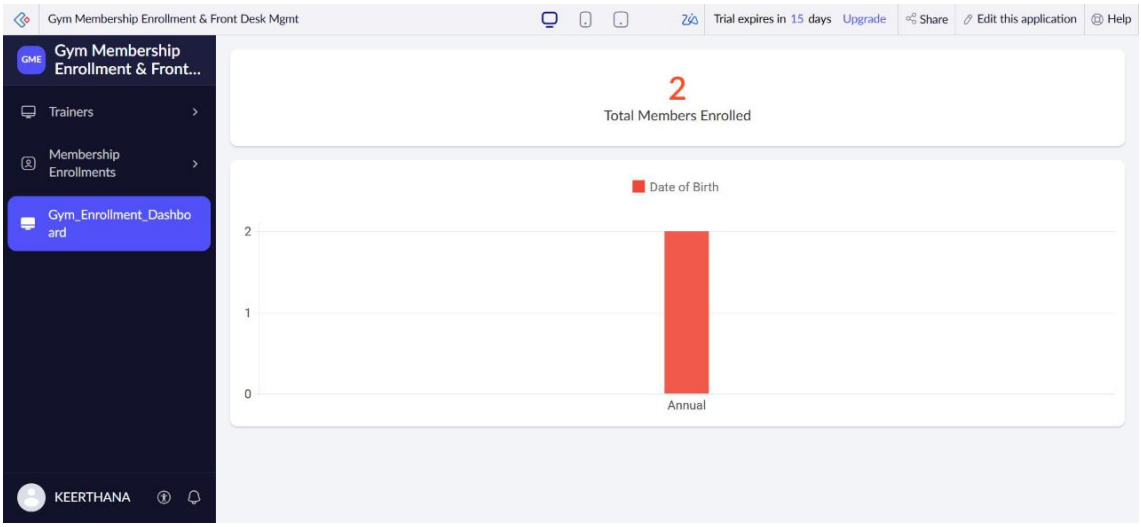


Fig 10.3 Dashboard summarising data

Result

Best practices were applied to the forms, report, workflow and dashboard.